

Traffic Accident Risk Analysis and High-Risk Zone Identification Using Data Analytics

Contents

- Business Objective
- Business Constraints
- Project Architecture - Data Flow Diagram
- Data Collection
- Exploratory Data Analysis
- Data Visualization

Project Overview and Scope

Project Overview

- Analyzed historical traffic data to understand congestion patterns, accident trends, and road safety risks.
- Identified high-risk locations, peak traffic hours, and violation behaviour across different road types and conditions.
- Performed data cleaning and feature engineering to create traffic level categories and risk indicators.

Project Scope

- Data cleaning and preprocessing using SQL.
- Exploratory Data Analysis (EDA).
- KPI calculation and statistical analysis.
- Interactive dashboards using Power BI.

Business Problem

Business Problem

- High number of accidents occurring at intersections and highways.
- Unpredictable traffic congestion during peak hours.
- Poor signal timing and infrastructure load imbalance.
- No data-driven system exists to identify accident-prone zones.
- Manual monitoring leads to delayed decision-making.

Business Objective

- Identify high-risk traffic zones using historical traffic patterns.
- Analyze congestion behaviour across time, weather and infrastructure.
- Enable proactive safety planning and resource allocation.

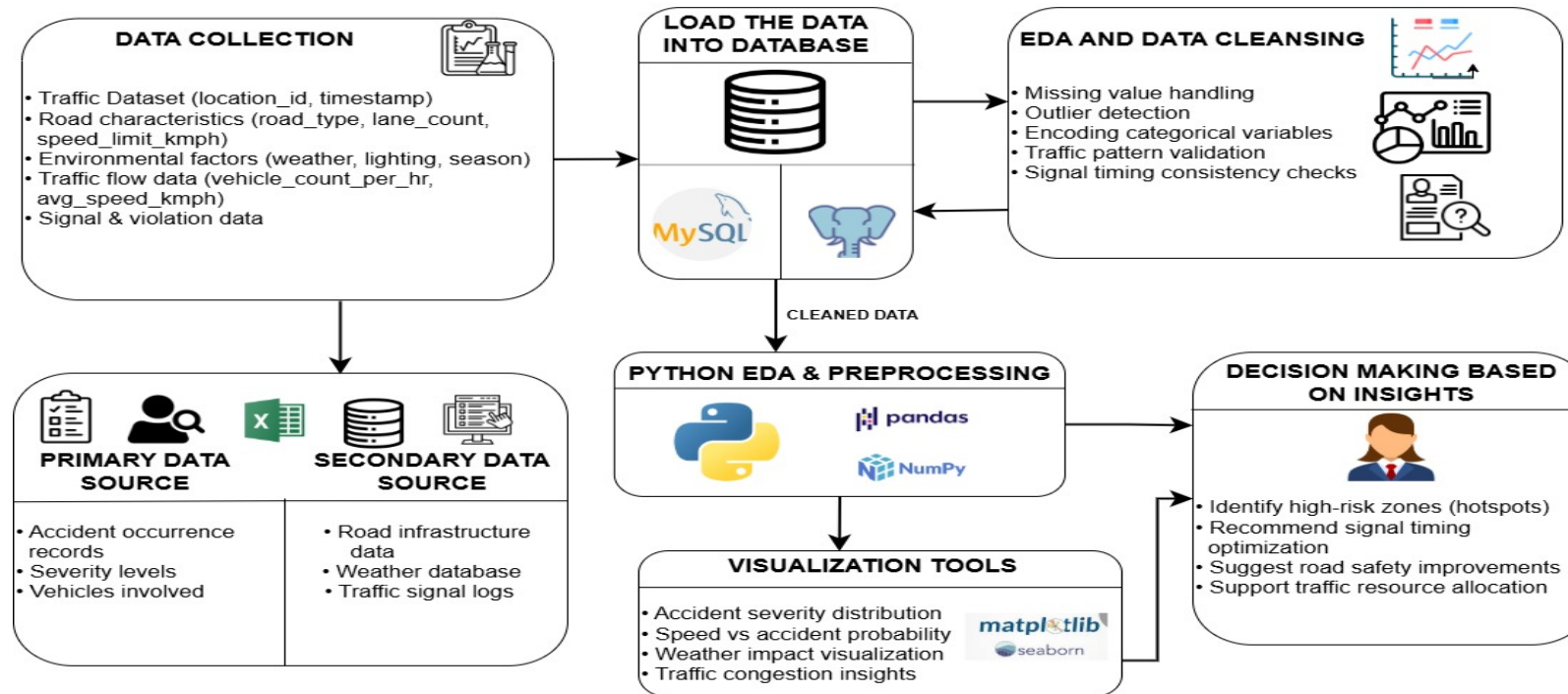
Business Constraint

- Inconsistent traffic signal and violation data.
- Weather and seasonal variability impacting traffic behaviour.
- Limited real-time monitoring capability.

Project Architecture – Data Flow Diagram

- The project follows a pipeline where raw traffic data is cleaned and transformed using Python and SQL, followed by feature engineering and visualization through interactive Power BI dashboards for decision support.

Project Architecture - Traffic Accident Risk Analysis & High-Risk Zone Identification



Data Collection

- Dataset designed based on real traffic monitoring parameters.
 - Includes traffic volume, speed limits, violations, accidents, lighting & weather.
 - Location-wise historical traffic patterns captured.
 - Infrastructure variables like lane count & signal timing included.
-

Data Dictionary

Column Name	Description	Data Type	Range
location_id	Unique identifier of traffic location/intersection	Categorical (String)	L001 to Ln
timestamp	Date and time of traffic observation	DateTime	Full timeline
state	State where traffic data is recorded	Categorical	Telangana
road_type	Type of road (urban/highway/rural)	Categorical	urban/highway/rural
lane_count	Number of lanes at location	Numeric (Integer)	1–10+
speed_limit_kmph	Maximum allowed speed on road	Numeric	40–120 kmph
has_signal	Whether signal present or not	Boolean	TRUE/FALSE
enforcement_level	Traffic enforcement strictness level	Categorical	low/medium/high
blackspot_score	Risk score of accident-prone zone	Numeric (Float)	0–1 or higher
latitude	Geographic latitude of location	Numeric (Float)	-90 to 90
longitude	Geographic longitude of location	Numeric (Float)	-180 to 180
season	Season during observation	Categorical	Summer/Monsoon/Winter
day_of_week	Day index of week	Numeric (Integer)	0 - 6
hour_of_day	Hour of traffic recording	Numeric	0–23
lighting	Lighting condition (Day/Night/Dusk)	Categorical	Day/Night/Dusk
weather	Weather condition	Categorical	Rain/Clear/Thunderstorm
is_peak	Whether peak traffic hour	Boolean	TRUE/FALSE
vehicle_count_per_hr	Number of vehicles per hour	Numeric	0–2000+
avg_speed_kmph	Average vehicle speed	Numeric	0–120
peak	Peak traffic indicator (duplicate flag)	Boolean	TRUE/FALSE

Exploratory Data Analysis [EDA]

Statistical Insights

- Vehicle count per hour shows high variability, indicating fluctuating congestion patterns across time periods.
- Average speed distribution is moderately skewed, with lower speeds observed during peak traffic hours.
- Violations and accident occurrences are sparse but concentrated in specific time and location clusters.
- Traffic signal cycle times and lane counts show noticeable variation across road types.
- Blackspot risk scores indicate uneven risk distribution across locations.

Business Insights

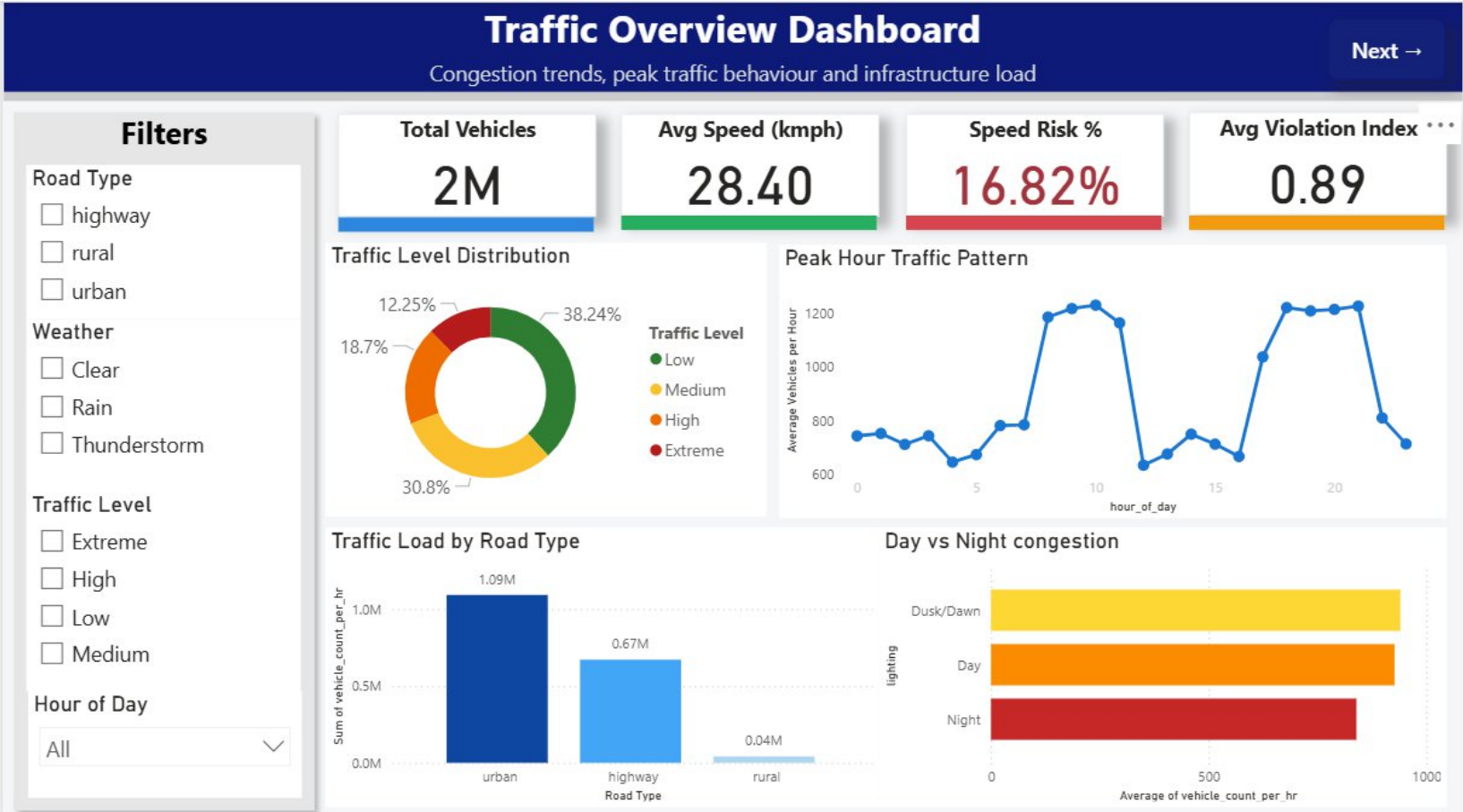
- Peak hour congestion trends highlight the need for dynamic traffic signal optimization.
- High-risk zones identified through blackspot scores can support targeted safety interventions.
- Violation patterns suggest increased enforcement requirements during specific weather and lighting conditions.
- Infrastructure differences between urban, highway, and rural roads influence traffic load and accident probability.
- Consistent monitoring of speed behaviour can help reduce accident risk and improve traffic flow efficiency.

Data Preprocessing

- Handled missing, null, and inconsistent values.
- Standardized timestamp format and extracted hour, day, and month features.
- Converted data types for speed, vehicle count, signal duration and violation metrics.
- Removed duplicate and invalid records.
- Validated latitude–longitude ranges and road type categories.
- Created analytical flags such as peak hour indicator and overspeed risk flag.
- Generated derived KPI fields like traffic level and risk zone classification.

Data Visualization

Traffic Overview Dashboard



- Monitors overall traffic congestion trends across locations and time periods.
- Identifies peak traffic hours to support signal timing and route planning decisions
- Compares traffic load distribution across road types like urban, highway, and rural.
- Helps traffic authorities optimize infrastructure utilization and reduce congestion risk.

Data Visualization

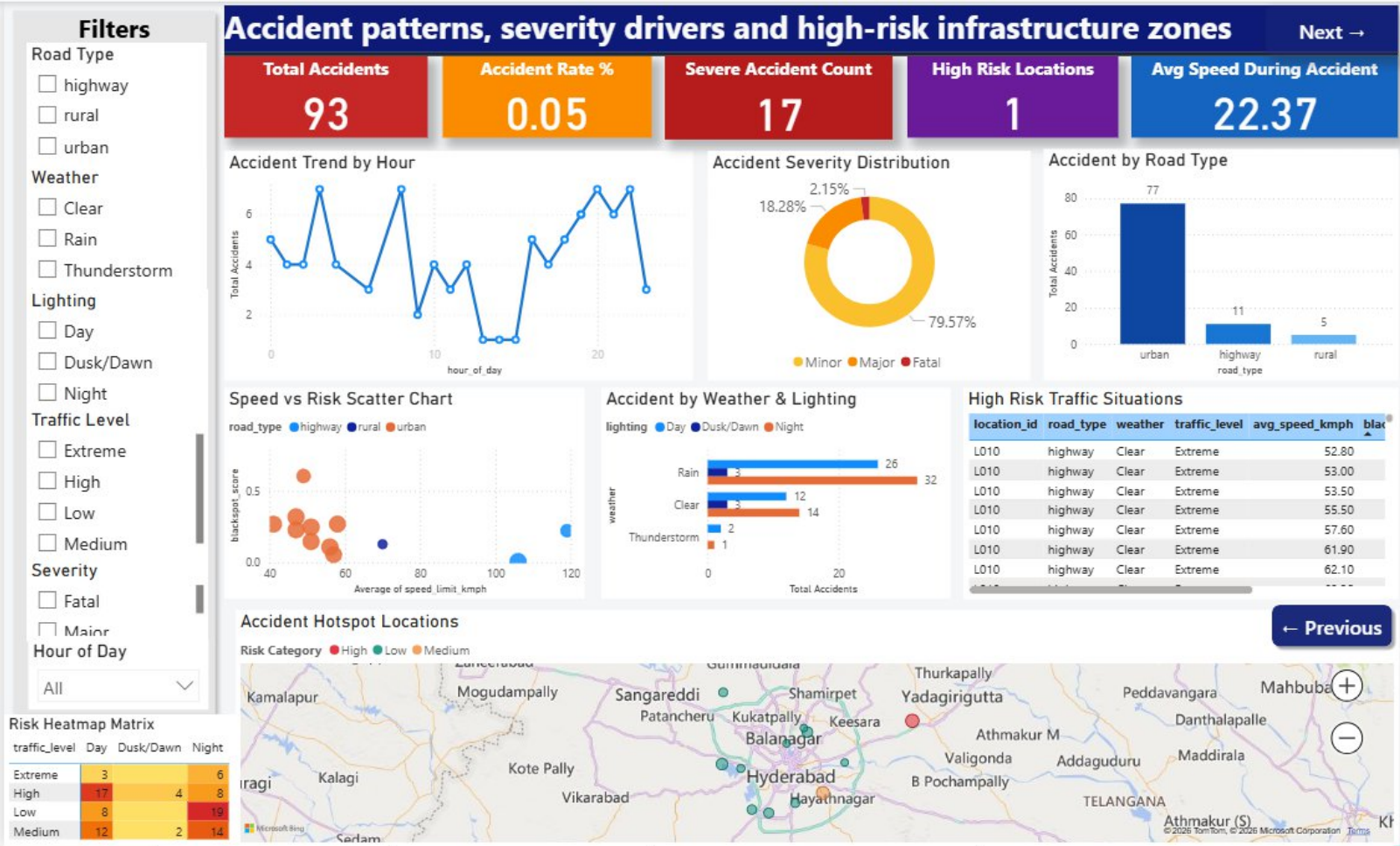
Safety & Violations Dashboard



- Tracks speed risk behaviour and violation intensity across different time intervals.
- Highlights high-risk combinations of weather, lighting, and traffic levels.
- Enables targeted enforcement planning in urban and high-violation corridors.
- Supports proactive safety monitoring to reduce accident probability.

Data Visualization

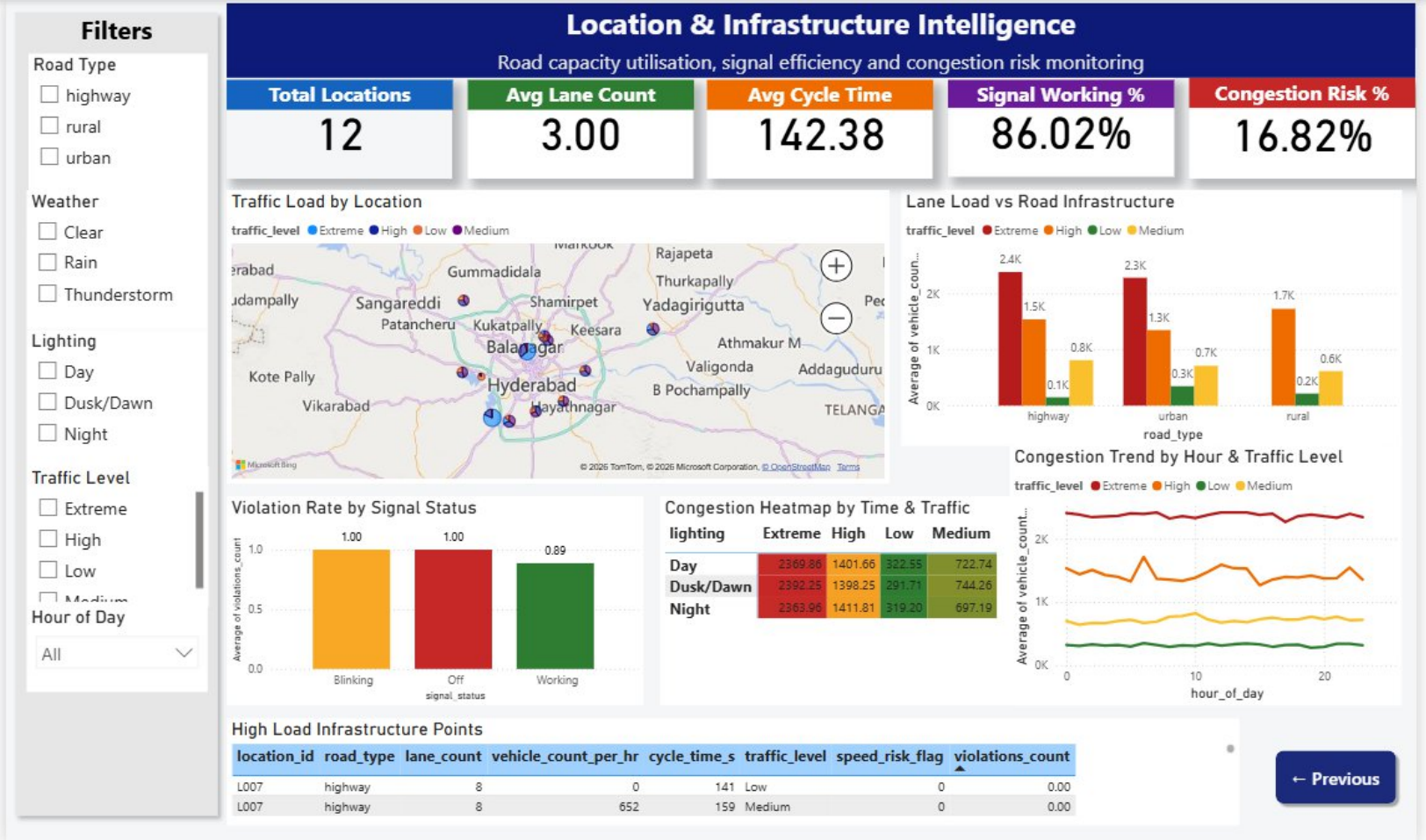
Accident & Risk Analysis Dashboard



- Analyzes accident severity distribution and hourly accident patterns.
- Identifies accident-prone road types and environmental risk factors.
- Detects high-risk traffic situations for faster emergency response planning.
- Supports data-driven safety interventions in critical locations.

Data Visualization

Location & Infrastructure Dashboard



Video

